

CSS + JS

Dark Mode Toggle

A Complete Classroom Lesson

Grade 4 · Web Development for Kids

HTML

CSS

JavaScript

DOM

Events

■ Lesson Overview

In this lesson, students will discover how the dark mode feature — found in apps like WhatsApp, YouTube, Zomato, BYJU'S, Swiggy, and Flipkart — actually works under the hood. Using three languages of the web (HTML, CSS, and JavaScript), students will build a fully functional dark mode toggle from scratch.

Duration	60–90 minutes
Grade Level	Grade 4 (Ages 9–10)
Subject	Computer Science / Web Development
Prerequisites	Basic familiarity with a code editor; no prior HTML/CSS/JS required
Deliverable	A working index.html file with a dark/light mode toggle button

■ Learning Objectives

- Explain what dark mode is and why it is used in modern apps.
- Identify the roles of HTML, CSS, and JavaScript in a webpage.
- Write a CSS class (*.dark-mode*) that changes background and text colors.
- Use *addEventListener()* to respond to a button click.
- Apply *classList.toggle()* to switch a CSS class on and off.
- Customize dark mode colors to create a personal design.

1.1 Understanding Dark Mode

Dark mode is a display setting where the screen uses a **dark background** combined with **light-coloured text and icons**. Instead of the usual white background with black text, the colours are flipped so that it is easier on the eyes — especially in dim environments like a bedroom at night.

Quick Definition

Dark Mode = Dark background + Light text

Light Mode = Light background (e.g. white #ffffff) + Dark text (e.g. near-black #121212)

1.2 Popular Apps That Use Dark Mode

Many apps that students use every day support dark mode. Here are some well-known Indian examples:

App	Why Dark Mode Helps
WhatsApp	Long chats at night — less strain on the eyes
YouTube	Videos look more vivid against a dark background
Zomato	Browsing menus late at night is more comfortable
BYJU'S	Reading textbook content in low light
Swiggy	Ordering food late at night without glare
Flipkart	Shopping in bed without a bright screen waking others

1.3 Benefits of Dark Mode

- **Saves Battery** — On OLED screens (like most modern phones), dark pixels use less electricity, so the battery lasts longer.
- **Reduces Eye Strain** — In low-light environments, a bright white screen can feel harsh. Dark mode reduces glare.
- **Looks Modern & Cool** — Many people simply prefer the sleek, professional look of dark mode!

■ **Tip:** Ask the class: How many of you use dark mode on your phone? What is your favourite app in dark mode?

2.1 The Chai Stall Analogy

Imagine a colourful **chai stall** you might see on any Indian street. It has three parts that work together — just like a webpage:

 HTML The chai stall cart itself — the structure that holds everything together. Without HTML there is nothing to look at!	 CSS Decorations and paint on the cart — makes everything look colourful and stylish. CSS is all about appearance.	 JavaScript The person making chai and taking orders — adds behaviour and action. JS makes the page do things when you click!
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2.2 Each Language in Detail

HTML — HyperText Markup Language

HTML is the skeleton of every webpage. It uses **tags** (words inside angle brackets like `<h1>` or `<p>`) to create structure. Think of it like labelling boxes: one box is a heading, another is a paragraph, another is a button.

CSS — Cascading Style Sheets

CSS controls how HTML elements **look**. You can set colours, font sizes, spacing, borders, and much more. In this lesson, CSS is what actually creates the dark and light colour schemes — JS just tells CSS which one to show.

JavaScript (JS)

JavaScript makes webpages **interactive**. When a user clicks a button, JS can respond — it can show/hide things, change text, fetch data from the internet, and in our case, switch between light and dark mode.

Analogy Summary

HTML = Body of a house | CSS = Paint & furniture | JS = Electricity & switches

3.1 What is a CSS Class?

A **CSS class** is a named set of style rules. Any HTML element can wear one or more classes — and when it does, it automatically follows all the style rules for that class. Adding or removing a class instantly changes how the element looks.

■ School Uniform Badge Analogy

Think of a school badge: students who wear it must follow special rules (uniform, behaviour). If you put the badge on, you follow the rules. Take it off, and you go back to normal. A CSS class works exactly the same way — add the class, the element follows those style rules; remove it, and the element returns to its default look.

3.2 The .dark-mode Class

We create a CSS class called **.dark-mode**. Whenever this class is added to the `<body>` element, the whole page switches to dark colours. Here is what it looks like in CSS:

```
/* ■■ Light Mode (default) ■■ */
body {
  background-color: #ffffff; /* white background */
  color: #121212; /* near-black text */
  transition: background-color 0.3s ease, color 0.3s ease;
}

/* ■■ Dark Mode ■■ */
.dark-mode {
  background-color: #121212; /* very dark background */
  color: #f0f0f0; /* off-white text */
}

.dark-mode a {
  color: #6200ea; /* purple links in dark mode */
}
```

3.3 Smooth Transitions

The **transition** property tells the browser to animate the change in colours over 0.3 seconds instead of switching instantly. This makes the dark mode feel polished and professional — just like real apps do it.

transition: 0.3s ease;

0.3s = 300 milliseconds (about one-third of a second). 'ease' means the animation starts slow, speeds up, then slows down again — creating a natural feel.

4.1 What is an Event?

In JavaScript, an **event** is something that happens in the browser — a user clicks a button, moves the mouse, presses a key, or the page finishes loading. Events let us write code that reacts to what the user does.

■ Doorbell Analogy

A doorbell listens all day. The moment someone presses it, it rings. JavaScript's `addEventListener()` does the same thing — it 'listens' for a specific event (like a click) and then 'rings' by running a function (your code).

4.2 `getElementById()`

Before we can attach an event listener, we need to find our button in the HTML. `getElementById()` searches the whole page and returns the element whose id matches the text we provide. We save this in a **variable** so we can use it later.

```
// Step 1: Find the button
const btn = document.getElementById("toggle-btn");
// 'btn' now holds a reference to our button element
```

4.3 `addEventListener()`

Once we have the button stored in `btn`, we attach a click listener. The first argument is the type of event ("click"), and the second is the function to run:

```
// Step 2: Listen for clicks
btn.addEventListener("click", function() {
  // This code runs every time the button is clicked
  document.body.classList.toggle("dark-mode");
});
```

5.1 The Ceiling Fan Switch Analogy

■ Ceiling Fan Switch Analogy

Think of the ceiling fan switch at home. Press it once → fan turns ON. Press it again → fan turns OFF. That is exactly what `classList.toggle()` does with a CSS class: call it once → class is ADDED. Call it again → class is REMOVED. Back and forth, every time.

5.2 How `classList.toggle()` Works

- **IF the class is NOT present** → It ADDS the class (dark mode turns ON)
- **IF the class IS already present** → It REMOVES the class (dark mode turns OFF)
- **Result:** Every click flips between light and dark mode!

```
// The magic one-liner:
document.body.classList.toggle("dark-mode");

// Equivalent to doing this manually:
if (document.body.classList.contains("dark-mode")) {
  document.body.classList.remove("dark-mode");
} else {
  document.body.classList.add("dark-mode");
}
```

■ *Tip: `classList.toggle()` is so useful because it saves us from writing the if/else every time. Real developers use it all the time!*

6.1 HTML Structure (index.html)

Type the following code into a new file saved as **index.html**. Every line matters — be careful with spelling and brackets!

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>My Dark Mode Page</title>
<link rel='stylesheet' href='style.css'>
</head>
<body>
<h1>Hello, Coder!</h1>
<p>Welcome to my webpage. Try toggling the theme!</p>
<button id="toggle-btn">■ Toggle Dark Mode</button>
<script src='script.js'></script>
</body>
</html>
```

6.2 CSS Styles (style.css)

```

/* ■■ Default (Light Mode) ■■ */
body {
font-family: Arial, sans-serif;
background-color: #ffffff;
color: #121212;
transition: background-color 0.3s ease, color 0.3s ease;
padding: 40px;
}
button {
padding: 10px 20px;
font-size: 16px;
cursor: pointer;
border: 2px solid #121212;
border-radius: 8px;
background-color: #f0f0f0;
}
/* ■■ Dark Mode ■■ */
.dark-mode {
background-color: #121212;
color: #f0f0f0;
}
.dark-mode button {
background-color: #333333;
color: #f0f0f0;
border-color: #f0f0f0;
}
.dark-mode a {
color: #6200ea; /* BYJU'S purple */
}

```

6.3 JavaScript Logic (script.js)

```
// Step 1: Find the toggle button
const btn = document.getElementById("toggle-btn");
// Step 2: Listen for a click event on the button
btn.addEventListener("click", function() {
// Step 3: Toggle the .dark-mode class on the body
document.body.classList.toggle("dark-mode");
});
```

7.1 Running Your Project

Once all three files (`index.html`, `style.css`, `script.js`) are saved in the same folder:

- 1. Open **index.html** in your browser (double-click the file, or click Run in your code editor).
- 2. You should see a white page with 'Hello, Coder!' and a toggle button.
- 3. Click the button — the background should turn dark and the text should turn light!
- 4. Click it again — it should switch back to light mode.
- 5. Keep clicking and watch the magic happen!

■ **Tip:** *If nothing happens when you click, open the browser's DevTools (F12) and check the Console tab for error messages.*

7.2 Personalise Your Dark Mode Colors

Now that the basics work, let's make it your own! Open **style.css** and try changing the colour values inside `.dark-mode {}`. Some fun ideas inspired by Indian apps:

App	Colour	Hex Code	Use it for
BYJU'S	Purple	#6200ea	Links or headings in dark mode
Swiggy	Orange	#fc8019	Button background or accent
Zomato	Red	#e23744	Headings in dark mode
WhatsApp	Teal	#00a884	Button border or icon colour
YouTube	Red	#ff0000	Accent colour on dark background

HTML

HyperText Markup Language. The language used to create the structure of a webpage using tags like `<h1>`, `<p>`, and `<button>`.

CSS

Cascading Style Sheets. The language used to describe how HTML elements look — colours, fonts, spacing, etc.

JavaScript (JS)

A programming language that runs in the browser and makes webpages interactive. It can respond to user actions.

Class (CSS)

A named group of CSS style rules. Elements can adopt a class to change their appearance instantly.

`getElementById()`

A JavaScript method that searches the page for an element with a specific id and returns it.

`addEventListener()`

A JavaScript method that 'listens' for a specific event (e.g. 'click') and runs a function when that event happens.

`classList.toggle()`

A JavaScript method that adds a CSS class if it isn't there, or removes it if it is — flipping back and forth with each call.

transition

A CSS property that animates a change in style smoothly over a set time (e.g. 0.3s ease).

DOM

Document Object Model. The browser's in-memory representation of the webpage, which JavaScript can read and modify.

Variable (`const`)

A named storage location in JavaScript. `'const btn = ...'` stores our button so we can use it later.

Event

Something that happens in the browser — a click, a key press, the page loading. JS can react to events.

Dark Mode

A colour scheme using a dark background and light text, reducing eye strain and saving battery.

9.1 Multiple Choice Questions

1. What does CSS stand for?

- A) Computer Style Sheets
- B) Cascading Style Sheets
- C) Colorful Screen Settings
- D) Creative Script System

Answer: B

2. Which language adds interactivity and behaviour to a webpage?

- A) HTML
- B) CSS
- C) JavaScript
- D) Python

Answer: C

3. What does `classList.toggle('dark-mode')` do?

- A) Always adds the dark-mode class
- B) Always removes the dark-mode class
- C) Adds the class if absent, removes it if present
- D) Changes the page font

Answer: C

4. What is the background-color hex code used for dark mode in this lesson?

- A) #ffffff
- B) #f0f0f0
- C) #6200ea
- D) #121212

Answer: D

5. What does `addEventListener('click', function)` do?

- A) Creates a new button
- B) Runs the function when the element is clicked
- C) Removes the button from the page
- D) Changes the button colour

Answer: B

9.2 Discussion Questions

- Which of the three web languages is most like an artist? Why?
- Can you think of another analogy (besides a doorbell) for `addEventListener()`?
- What happens if you call `classList.toggle('dark-mode')` five times in a row? Will dark mode be ON or OFF?
- What other CSS properties could you change in dark mode? (Hint: think of font-size, border-color, box-shadow...)
- Why might a school website want a dark mode option?

For students who finish early or want an extra challenge:

■ Level 1 — Change the Button Text

Make the button say 'Switch to Dark Mode' when in light mode and 'Switch to Light Mode' when in dark mode. Hint: check if `document.body.classList.contains('dark-mode')` and update `btn.textContent`.

■■ Level 2 — Remember the Choice

Use `localStorage.setItem()` to save whether dark mode is on. When the page loads, read the saved value with `localStorage.getItem()` and apply the class automatically.

■■■ Level 3 — Animated Sun/Moon Icon

Add a moon emoji 🌙 and sun emoji ☀️. Show the moon when dark mode is off and the sun when it is on. Use CSS transition to animate the switch.

■■■ Level 4 — Full Dark Mode Website

Build a mini website with a navigation bar, an image, and a footer. Apply dark mode styles to all elements. Try using CSS variables (`--bg-color`, `--text-color`) so you only need to update values in one place.

■ Teacher Notes & Tips

- Have students open the browser DevTools (F12 → Elements) and watch the class appear and disappear on `<body>` as they click the button. This makes the abstract concept very visual.
- Encourage students to experiment with colours freely — there are no wrong choices. The goal is to make them feel like real developers.
- Pair slower students with faster ones during the typing activity. The driver/navigator pair programming model works well here.
- If students are using an online editor (e.g. CodePen, Replit), they can put all three sections in one file — CSS in a `<style>` block and JS in a `<script>` block inside `index.html`.
- Consider projecting your screen while typing the code so students can follow along step by step.
- The transition CSS property is optional but highly recommended — students are always delighted by the smooth colour animation.

Happy Coding! ■ See you in the next lesson.